

PULSE — Regime-Conditional Crude Spread Engine

Methodology & Walk-Forward Results · Generated 2026-06-15 03:11 UTC · Reproduce: `python -m backend.research.walkforward` · `python -m backend.research.methodology_pdf`

1. Problem

Identify when crude calendar spreads (front carry, mid-curve carry, front butterfly) deviate from their *regime-conditional* fair value. Output a single ranked opportunity per day with full receipts: fair, 80% band, residual z-score, top drivers, historical analogs.

2. Universe

6 instruments — 3 Brent (M1-M2, M3-M6, front fly M1-2×M2+M3) + 3 WTI mirror instruments. Brent legs from /Data C1-C31 daily settlements (2016-2026, 2,712 trading days). WTI legs from synthesised daily settlements (last 1-min mid per session, 2021-2026, 1,676 trading days — flagged ESTIMATE pending the real exchange file).

3. Regime grids — composite + pooled

Composite (Sprint 3): 27 cells = **CURVE** {CONTANGO / NEUTRAL / BACK on Brent M1-M12 at -\$2, +\$5} × **INVENTORY** {LOW / AVG / HIGH on US crude stocks vs 5y seasonal at ±4%} × **VOL** {CALM / NORMAL / STRESSED on Brent 20d realised vol at 20%, 35%}. Hard thresholds — mentor can answer ‘why this regime?’ in one line. ~10-15 cells/spread have usable n ; the rest are economically implausible.

Pooled (Phase 2.5): 3 cells/spread = curve axis only {CONTANGO / NEUTRAL / BACK}. Same thresholds; inventory + vol still surface on the UI as regime context but no longer fragment the training set. Average ~770 rows per cell (vs ~150 composite) — explicitly addresses the Sprint 4 finding that per-cell data was too thin to beat a 252-d rolling z .

4. Feature set (Phase 2.8.2 expansion)

20 Brent features per cell (was 11): *Curve shape* — M1-M12, its square, and *curvature* (M1 – 2×M6 + M12) for fly-specific information. *Front state* — Brent close + 5d return + 20d realised vol. *Inventory* — US crude vs 5y seasonal AND **inventory surprise** (weekly Δ minus its 4-week MA). *Seasonality* — sin/cos of day-of-year. *Lagged spreads* — M1-M2 / M3-M6 / fly lag-1.

Phase 2.8.2 alpha features (new) — **COT managed-money percentile** (rolling 156-week, contrarian crowding), **3-2-1 crack + gasoline crack** (refining-margin pressure on the front), **WTI-Brent spread** (Atlantic-basin arb), **real rate** (DGS10 – 5y TIPS breakeven; storage-cost driver), **OVX/VIX ratio** (crude vs equity vol risk-premium), and **days to front-month expiry**. WTI cells append 5 more (WTI close, M1-M12, 3 lag-1 spreads). The target spread’s own lag is removed to avoid self-leakage.

5. Per-cell model competition

We *do not* pick a single regression family a priori. For each (spread, regime) cell, **seven candidates** compete on 5-fold TimeSeriesSplit cross-validated R^2 (Phase 2.8.1 expansion):

Linear: Ridge (shrinks coefficients — good with correlated predictors), Lasso (zeroes out noise features — good when 3-4 signals dominate), ElasticNet (L1+L2 blend — mixed correlation), Huber (robust loss — COVID-2020 / Russia-2022 outliers).

Gradient boosting (Phase 2.8.1): XGBoost, LightGBM, CatBoost — shallow trees with strong regularisation that capture non-linear interactions between features. Win in cells where linear leaves alpha on the table.

Winner = max mean CV R^2 , simplicity tiebreak within 0.005: linear families rank above boosters (ElasticNet > Lasso > Ridge > Huber > XGBoost > LightGBM > CatBoost), so interpretability wins when CV scores are close. Quantile regressors (p10/p50/p90) are fit separately for the 80% confidence band.

6. Trading rule

At each daily settle: classify today’s composite regime → look up winner model for (spread, regime) → predict fair, p10, p50, p90 → $z = (\text{actual} - \text{fair}) / \text{residual}_\sigma$ from training. Direction = SELL if $z > +0.5\sigma$, BUY if $z < -0.5\sigma$, else NEUTRAL. Target = p50 (mean reversion). Stop = entry $\pm 1.5\sigma$. Horizon = 20 trading days.

7. Walk-forward design

Refits every quarter from 2024-Q1 through 2026-Q2 (34 refits). Each refit re-runs the full 4-model competition on data $\leq$ cutoff, then we walk forward day-by-day to the next cutoff, generating signals. **Phase 2.5 runs BOTH grids in parallel** (composite + pooled) over the same refit dates and the same trading rule, then compares each to a regime-unaware 252-day rolling z -score baseline — isolating the contribution of regime conditioning and of grid resolution.

8. Phase 2.6 — gated blend (production rule)

Phase 2.5 surfaced the BACK × {Lasso, Huber} slice as the only (spread × regime × winner) subset where the pooled engine demonstrably beats baseline in walk-forward. **Phase 2.6 codifies that finding as a production rule** and tests it as a third leg alongside composite / pooled / baseline. For each (date, spread): (i) if the pooled candidate has *regime_pooled*='BACK', *winner_model* ∈ {Lasso, Huber}, AND $|z| \geq 0.5\sigma$ → take the pooled signal; (ii) else fall back to the 252-day rolling- z baseline. This routing is the exact rule shipped under PULSE_GATED_BLEND=1, so the table on page 2 measures the strategy a trader would actually run.

9. Phase 2.7 — position sizing for the regime leg

Phase 2.6 left max drawdown on the gated blend at -271 (vs -169 baseline) — the cost of concentrating 97 high-Sharpe regime fires on a small number of days. **Phase 2.7 scales the regime-leg notional** while leaving the baseline leg at 1.0, asking whether sizing compresses the DD without giving up the +1.332 regime-leg Sharpe. Three modes simulated end-to-end as a fourth walk-forward leg behind PULSE_GATED_SIZE=<mode>:

full — scale 1.0 (sanity check; identical to gated_blend). **half** — scale 0.5 (uniform risk reduction). **kelly** — per-spread Kelly fraction $f^* = p - (1-p)/b$ computed at each refit boundary from prior closed regime-leg PnLs (p = win rate, b = mean_win / |mean_loss|). Clamped to [0.10, 1.00]; defaults to 0.50 when fewer than 5 prior trades exist. The schedule is expanding-window so the per-spread sample grows over time. Live inference reads the per-spread Kelly at the latest refit boundary from sized_blend_summary.kelly_per_spread_latest.

10. Phase 2.8 — real model + real features

Phase 2.8.1 adds **XGBoost / LightGBM / CatBoost** to the per-cell competition so cells with non-linear interactions aren't forced to fit a linear model. Boosters fit only on cells with ≥ 80 train rows (smaller samples can't reliably distinguish trees from linears via CV); shallow trees (depth 3-4) and low learning rate (0.05) keep them from memorising. Boosters rank below linears in the tiebreak so interpretability still wins on ties.

Phase 2.8.2 nearly **doubles the feature set** (11 $\rightarrow$ 20 Brent features) with the alpha-bearing predictors documented in §4: COT crowding, inventory surprise, curvature, refining cracks, real rate, OVX/VIX ratio, WTI-Brent arb, days-to-expiry. The aim is to give every model family — linear and tree — a richer set of predictors so the gated-blend Sharpe lifts by tightening the pooled BACK fair-value estimates the gate rule relies on.

Phase 2.8.3 widened the gate to admit boosters (GATED_WINNERS = {Lasso, Huber, XGBoost, LightGBM, CatBoost}). The widened gate is methodologically correct — the regime leg's reported Sharpe lifted from +0.369 to **+0.888** across 244 fires (was 71) — but the headline blended Sharpe stayed flat (+0.389 $\rightarrow$ +0.384) because most of the newly-routed booster trades had baseline-z direction matching pooled-z direction. The widening REASSIGNS attribution rather than CREATING new alpha at the blended level.

Phase 2.8.4 — **global model with regime-as-feature**. The 2018-2026 walk-forward (Phase 2.8.8) showed the regime-unaware baseline beats every per-cell variant on NET Sharpe — so the per-cell *split* is itself the problem. Phase 2.8.4 collapses the grid: **ONE** model per spread trained on *all* rows, with the composite regime fed as **9 one-hot axis columns** (curve / inventory / vol — 3+3+3) instead of partitioning the data. Same 7-model competition. Each spread now trains on $\sim 5\times$ more rows per refit, giving boosters a real shot at picking up regime $\times$ feature interactions the per-cell linear winners can't express. Reported alongside the other modes on page 2 — same NET cost model, same horizon, same 34 refits.

10b. Phase 2.8.5 — soft regime probabilities

Complementary test to 2.8.4. The hard pooled grid forces **discontinuous** regime assignment at -\$2 and +\$5 on M1-M12; two days at -1.9 and -2.1 flip cell, model, and residual scale even though the market is virtually identical. Phase 2.8.5 replaces the indicator with a **logistic membership function** at each threshold (bandwidth \$1/bbl for curve and inventory, 2.5pp for vol) — at the boundary a day is 50/50 across adjacent regimes, far from the boundary it collapses back to the hard label. Each day's prediction is the *posterior-weighted blend* across all trained cells for that spread (point, quantiles, and residual variance all blended). Bandwidths come from the trader thresholds, not tuned to PnL.

Honest framing: hard pooled recovers as bandwidth $\rightarrow 0$; a flat prior across cells recovers as bandwidth $\rightarrow \infty$, which is approximately the 2.8.4 global model. So 2.8.5 sits on the interpolation between 'hard split' and 'single global model' — the question it answers is whether a *small* amount of cell-sharing recovers alpha the discontinuous cut leaks. Reported on page 2 alongside the other modes under the same 34-refit walk-forward and the same NET cost model.

11. Phase 2.8.6 — transaction costs

Every Sharpe/PnL number above is **gross** — per-trade transaction cost is not modelled by the walk-forward's spread-move PnL. Phase 2.8.6 layers a defensible cost model on top of the existing trade tape so the table on page 2 reports **NET (after-cost)** Sharpe alongside gross.

Cost model. Per-leg, per-side: \$0.0025/bbl commission + clearing + brokerage ($\sim$ \$2.45 per 1,000-bbl contract) + half bid-ask slippage (\$0.0050/bbl for front M1/M2, \$0.0075/bbl for deferred M3-M6). Round-trip cost = $N \text{ legs} \times 2 \text{ sides} \times (\text{commission} + \text{half-spread})$. That gives: **2-leg M1-M2 = \$0.030/bbl RT, 2-leg M3-M6 = \$0.040, 3-leg fly = \$0.050**. Cost scales with sizing_scale on regime rows (half-sized regime trade = half the contracts = half the fees). NEUTRAL trades incur zero cost (no fill).

Cost is applied at the aggregation step only — per-cell CV is on gross fair-value R^2 , so winner selection is unaffected and no retraining is required. Live inference does *not* subtract cost at entry; the cost-aware NET metric exists purely for backtest reporting.

Walk-forward results — 2024-2026

Refits: 34 · Records/mode: 8,421 · Horizon: 20 trading days · Universe: 6 spreads · Composite grid: 27 cells · Pooled grid: 3 curve cells · Gated blend (Phase 2.6): regime engine fires on BACK + {Lasso,Huber} + $|z| \geq 0.5\sigma$, else 252d z baseline (regime/baseline split: 1,567 / 8,379) · Sized blend (Phase 2.7): regime-leg notional scaled by {full=1.0, half=0.5, kelly=per-spread f^* }

Overall — composite / pooled / gated blend / baseline

Metric	Composite (27)	Pooled (3)	Gated blend	Baseline 252d z	Δ gated vs base
Signals fired	6380	7160	6598	6140	—
Hit rate	57.9%	59.4%	63.8%	65.2%	-1.4%
Mean PnL (\$/bbl)	+0.133	+0.142	+0.148	+0.177	-0.029
Median PnL	+0.070	+0.080	+0.130	+0.150	—
Total PnL	+846.2	+1013.5	+974.4	+1087.1	—
Sharpe (ann.)	+0.38	+0.41	+0.41	+0.48	-0.07
Max drawdown	-102.1	-125.3	-275.7	-168.9	—

Phase 2.6 finding: even the gated blend can't pull the headline above baseline — the BACK + {Lasso,Huber} slice was strong in Phase 2.5 but it didn't survive a clean walk-forward of the gating rule itself. Sharpe (ann.): composite +0.38, pooled +0.41, **gated +0.41**, baseline +0.48. Gate routing: 1,567 regime fires / 8,379 baseline fallbacks (15.8% of records hit the engine). The gate doesn't beat baseline in walk-forward. Next: try a wider feature set inside the pooled regression (EIA inventory surprise, COT positioning, OVX, term-structure derivatives), or revisit gating criteria — the Phase 2.5 slice may have been window-specific.

Per-spread Sharpe — composite / pooled / gated / baseline (Δ vs baseline)

Spread	Comp Shp	Pool Shp	Gate Sig	Gate Shp	Base Shp	Δ Gate
brent_fly_123	+1.11	+1.02	1106	+1.28	+1.52	-0.24
brent_m1_m2	+0.70	+0.61	1490	+0.71	+0.77	-0.06
brent_m3_m6	-0.15	+0.05	1619	+0.09	+0.06	+0.02
wti_fly_123	+0.76	+0.91	731	+0.86	+0.97	-0.11
wti_m1_m2	+0.54	+0.33	817	+0.33	+0.33	+0.00
wti_m3_m6	+0.31	+0.36	835	+0.10	+0.39	-0.29

Gated-blend slices — which leg fired (regime vs baseline) and winner mix

Gated-blend slice	n	Hit	μ PnL	Sharpe
source=baseline	5071	64.7%	+0.137	+0.38
source=regime	1527	60.7%	+0.184	+0.48
winner=CatBoost	584	82.9%	+0.346	+1.21
winner=Huber	1788	58.7%	+0.086	+0.32
winner=UNKNOWN	107	80.4%	+0.334	+3.26
winner=XGBoost	1636	60.8%	+0.147	+0.47
winner=Ridge	2483	64.2%	+0.138	+0.30

Phase 2.7 — sized regime leg vs gated full / baseline

Metric	Gated full	Sized 0.5×	Sized Kelly	Baseline 252d z	Δ best vs base
Signals fired	6598	6598	6598	6140	—
Hit rate	63.8%	63.8%	63.8%	65.2%	—

Mean PnL (\$/bbl)	+0.148	+0.126	+0.123	+0.177	—
Total PnL	+974.4	+833.8	+814.6	+1087.1	—
Sharpe (ann.)	+0.41	+0.39	+0.38	+0.48	-0.07
Max drawdown	-275.7	-295.2	-295.1	-168.9	-106.9
Mean regime scale	1.0	+0.500	+0.342	—	—

Phase 2.7 finding: no sizing mode caught baseline at the headline Sharpe (best gated full +0.41 vs baseline +0.48). Max DD on the best variant: -275.7 vs baseline -168.9 (gated full: -275.7). Latest per-spread Kelly fractions: brent_m1_m2=0.18, brent_m3_m6=0.10, brent_fly_123=0.37, wti_m1_m2=0.44, wti_m3_m6=0.10, wti_fly_123=0.51.

Phase 2.8.6 — NET headline (after transaction costs)

Cost model: per-leg per-side \$0.0025 commission + half-spread slippage (\$0.0050 front / \$0.0075 deferred). RT cost per spread: M1-M2 \$0.030, M3-M6 \$0.040, fly \$0.050.

Mode	Gross / Net Sharpe	Gross / Net μ PnL	Mean cost / fire	n fired
Baseline 252d z	+0.48 / +0.37	+0.177 / +0.138	+0.0389	6140
Pooled (un-gated)	+0.41 / +0.29	+0.142 / +0.101	+0.0404	7160
Gated blend (2.8.3)	+0.41 / +0.30	+0.148 / +0.108	+0.0393	6598
Global (Phase 2.8.4)	+0.49 / +0.38	+0.177 / +0.136	+0.0410	6263
Pooled-soft (2.8.5)	+0.42 / +0.30	+0.141 / +0.101	+0.0404	7310
Sized half	+0.39 / +0.28	+0.126 / +0.092	+0.0343	6598
Sized kelly	+0.38 / +0.28	+0.123 / +0.090	+0.0329	6598

Spread	Gross Shp (gated)	Net Shp (gated)	Net Shp (base)	Δ Net vs base
brent_fly_123	+1.28	+1.02	+1.26	-0.24
brent_m1_m2	+0.71	+0.61	+0.67	-0.06
brent_m3_m6	+0.09	-0.01	-0.04	+0.03
wti_fly_123	+0.86	+0.67	+0.80	-0.13
wti_m1_m2	+0.33	+0.26	+0.26	+0.00
wti_m3_m6	+0.10	+0.02	+0.31	-0.29

Phase 2.8.6 finding: transaction costs erode Sharpe roughly uniformly across modes — baseline drag +0.10, pooled +0.12, gated_blend +0.11. Mean cost per fire sits at \$0.039-\$0.045/bbl depending on spread mix (boosters fire more on the fly which is the most expensive class). **Best net Sharpe: baseline at +0.372.** Gated blend net Sharpe (+0.298) vs baseline net (+0.372) = Δ -0.074 — the gated blend loses to baseline at the net headline. Transaction costs widen the gap because the regime fires concentrate on the fly (highest per-trade cost class). **Phase 2.8.4 (global model):** net Sharpe +0.380 — ties baseline (+0.372). Regime-as-feature is not a headline lift on this universe, but it doesn't cost anything either — the per-cell *split* was the binding constraint. **Phase 2.8.5 (soft pooled):** net Sharpe +0.297 — softening the curve thresholds does not lift pooled past baseline (+0.372). Discontinuity at the trader cuts isn't the binding constraint on this universe (mean modal weight 0.86, ~2.5 cells blended per fire).

Caveats & next steps

- **WTI is synthesised** — last-1-min midprice per session, not exchange print. Mentor's real WTI C1-C3 daily file slots in without code changes.
- **Sparse cells (composite only)** — 96/162 composite cells (BACK $\times$ HIGH, CONTANGO $\times$ LOW) stay empty. The pooled grid is dense by construction (3 cells/spread, average ~770 rows/cell vs ~150 composite).
- **Phase 1 not directly comparable** — Phase 1 is a 9-indicator directional Brent signal; the baseline here (regime-unaware 252d z on spreads) is the cleanest test of regime-conditioning value.
- **Costs modelled (Phase 2.8.6)** — defensible per-spread RT cost (\$0.030 M1-M2 / \$0.040 M3-M6 / \$0.050 fly per bbl) layered on top of gross spread-move PnL. Roll slippage and funding still ignored; live deployment should add a per-spread broker quote where available.
- **Mode switch** — set PULSE_REGIME_MODE=pooled for un-gated pooled inference, PULSE_GATED_BLEND=1 (Phase 2.6) for the gated blend, and PULSE_GATED_SIZE=<full|half|kelly> (Phase 2.7) to scale the regime-leg notional. Default is composite + full sizing for back-compat with Sprint 3 dashboard cards.

- **Gate is conservative by design** — Phase 2.5 demonstrated lift only on (BACK regime × {Lasso, Huber} winners). The gated blend codifies exactly that slice; per-spread Sharpe and the regime/baseline split in the headline disclose how often the engine actually fires versus deferring to the baseline.
- **Phase 2.7 sizing has a Sharpe-vs-DD trade-off** — uniform 0.5× halves both mean PnL and max DD on the regime leg; Kelly is per-spread and adaptive but its sample is thin (97 regime fires total). Live deployment should read `sized_blend_summary.kelly_per_spread_latest` from the walk-forward report — the same numbers shown above — to size at inference time.
- **Next:** fold inventory + vol back in as *features* inside the pooled regression so they inform fair value without fragmenting the training set; trial 5/60-day horizons (one-line change in `walkforward.py`); consider per-spread gate thresholds where the per-spread lift table justifies them.